Practical 68%
Oral 23%

OBSERVASTION CHECKLIST

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY DEPARTMENT

MODULE 2, TERM 2 — UNIT: NETWORK SETUP

PRACTICAL 3 CHECKLIST

Unit Code: 0612 351 03A

ELEMENT 2:NETWORK INSTALLATION(NETWORK CABLE TESTING)

CANDIDATE AND ASSESSOR DETAILS

ASSESSOR NAME.....James.....

CANDIDATE NAME.....Collins Ngotho.....

ADM NO.....1083.....

ASSESSMENT VENUE.....Computer lab 210.....

1. Assessment Details

Particulars	Details
Institution	Aberdare Technical College
Department	Information Communication Technology
Unit Name	Network Setup

CamScanner

Unit Code	0612 351 03A
Module / Term	Module 2, Term 2
Date of Assessment	27/10/2025
Time Allocated	2 Hours

2. Cutting List / Materials and Tools

Item	Quantity	Provided by
Cat5e/Cat6 UTP cable, 2 m pieces	3 pieces	Institution
RJ45 connectors, with boots if available	6 pieces	Institution
Wire stripper and cutter	1 set	Institution
Crimping tool	1	Institution
Network cable tester with remote unit	1	Institution
Digital multimeter	1	Institution
Measuring tape	1	Institution
Rim paper / foolscap for sketching	1 sheet	Institution

CamScanner

3. Practical Observation Checklist — 50 Marks

Instructions to assessor:

Observe the candidate performing each task.

Award marks according to demonstrated performance and record comments where necessary.

Task 1: Cable Preparation Using the Wire Stripper and Cutter — 8 Marks

No.	Observation criteria	Marks	Candidate score	Comments
1	Measures and cuts the cable to exactly 1 meter	2	02	measured and cut 1 meter cable
2	Strips approximately 2.5 cm of outer jacket at both ends	2	02	2.5 Jacket stripped
3	Avoids nicking or damaging the inner conductors	2	02	Cable is not damaged
4	Untwists and straightens the conductors appropriately	2	02	untwists and straightens conductors
	Total	8	08	

Task 2: Crimping an RJ45 Straight-Through Cable — 10 Marks

No.	Observation criteria	Marks	Candidate score	Comments
1	Arranges conductors in the correct T568B colour sequence	4	04	conductors arranged using T568B
2	Trims the conductors evenly	1	01	conductors trimmed evenly
	Inserts all conductors fully into the RJ45 connector	2	02	Conductors inserted fully
	Uses the crimping tool correctly	2	01	Crimping tool used correctly
	Repeats termination correctly at the other end	1	01	Termination repeated correctly

	Total	10	09	

Task 3: Continuity Testing with a Multimeter — 8 Marks

No.	Observation criteria	Marks	Candidate score	Comments
1	Sets the multimeter to continuity or the lowest resistance range	2	02	multimeter set to continuity
2	Tests corresponding pins at both ends of the cable	2	02	Pins tested both ends
3	Records beep/resistance readings and continuity results for all eight pins	2	02	Records resistance
4	Tests different pins to check for short circuits and interprets the results	2	02	tested
	Total	8	08	

Task 4: Wire Mapping with a Cable Tester — 8 Marks

No.	Observation criteria	Marks	Candidate score	Comments
1	Connects both cable ends to the main and remote tester units	2	02	connected
2	Switches on the tester and observes the LED sequence	2	02	switched on
3	Records the observed LED sequence accurately	1	01	Records LED sequence
4	Sketches the pin-to-pin wire map correctly	2	02	sketched
5	States whether the cable passes or fails and identifies the cable type	1	01	stated Cable passed
	Total	8	08	

Task 5: Cable Length Testing — 6 Marks

No.	Observation criteria	Marks	Candidate score	Comments
1	Measures the cable using a measuring tape	2	00	measured but incorrect value
2	Records the tester's length reading, if available, or explains how a TDR-type tester measures length	2	~~00~~ 02	NOT tested
3	Compares the measurements and records the difference	1	00	NOT compared

CamScanner

4	States the maximum permitted Ethernet segment length	1	01	stated but incorrect
	Total	6	02	

Task 6: Fault Detection and Rectification — 10 Marks

No.	Observation criteria	Marks	Candidate score	Comments
1	Tests the faulty cable using the cable tester and multimeter	2	02	tested the faulty cable
2	Identifies the fault type and affected pins	2	01	
3	Records the fault findings accurately	1	01	
4	Rectifies the fault by cutting, stripping and re-crimping as required	3	03	rectifies the fault
5	Retests the cable and confirms whether it now passes	2	02	Retests the cable
	Total	10	09	

Practical Marks Summary

Task	Maximum marks	Candidate score
Task 1: Cable preparation	8	08
Task 2: RJ45 termination	10	09
Task 3: Continuity testing	8	08
Task 4: Wire mapping	8	08

CamScanner

Task 5: Length testing	6	8
Task 6: Fault rectification	10	9
TOTAL	50	

4. Product Checklist — 5 Marks

Instructions to assessor: Inspect the candidate's completed cable and recorded assessment outputs. Award 1 mark for each acceptable product.

No.	Product requirement	Marks	Candidate score	Comments
1	Completed straight-through cable with RJ45 connectors terminated at both ends	1	1	
2	Completed continuity observation table covering pins 1–8	1	1	
3	Completed wire-map sketch showing the connections between both ends	1	1	
4	Completed cable length measurement and comparison record	1	1	
5	Fault report documenting the identified fault and retest result	1	1	
	Total	5	4½	

CamScanner

5. Oral Assessment Checklist — 30 Marks

No.	Topic	Question	Expected Answer	Marks	Score	Remarks
1	2.3.1 Cable Testing Methods	What is network cable testing?	Checking a cable's physical and electrical condition to confirm it can carry data reliably.	1	1/2	Incomplete
2	2.3.1 Cable Testing Methods	Why is cable testing important?	Detects faults early, ensures performance, reduces downtime and confirms the installation meets standards.	1	0	Importance stated
3	2.3.1 Cable Testing Methods	Name any two cable testing methods.	Continuity testing, wire mapping, length testing, fault detection.	1	0	Not given
4	2.3.1 Cable Testing Methods	When should a cable be tested?	After installation or termination, during troubleshooting and during routine maintenance.	1	1/2	.
5	2.3.2 Continuity Testing	What is continuity testing?	Checking for an unbroken electrical path from one end of a wire to the other.	1		
6	2.3.2 Continuity Testing	What does a continuity test detect?	Open circuits (breaks) in the conductors.	1	0	
7	2.3.2 Continuity Testing	What does a failed continuity test indicate?	A broken wire, bad connection or poor termination.	1	0	Indicators stated
8	2.3.2 Continuity Testing	Which tool is commonly used for continuity testing?	A multimeter (continuity mode) or a cable tester.	1	0	
9	2.3.3 Wire Mapping	What is wire mapping?	Verifying that each wire is connected to the correct pin at both ends.	1	0/0	
10	2.3.3 Wire Mapping	Name two wiring faults detected by wire mapping.	Any two: open, short, crossed pairs, reversed pairs, split pairs.	1	0	faults named
1	2.3.3 Wire Mapping	What is a split pair?	Wires from different twisted pairs are wired as one pair, causing crosstalk even though continuity passes.	1		
	2.3.3 Wire Mapping	Which standards define the	T568A and T568B.	1	0	standards named

CamScanner

		Ethernet wiring order?				
13	2.3.4 Cable Length Testing	Why do we test cable length?	To ensure the cable does not exceed the maximum distance, beyond which the signal weakens.	1	0/1	
14	2.3.4 Cable Length Testing	What is the maximum length of a UTP Ethernet channel segment?	100 metres (90 m permanent link plus patch cords).	1	1	explained
15	2.3.4 Cable Length Testing	Which technique measures cable length?	Time Domain Reflectometry (TDR).	1	1	
16	2.3.4 Cable Length Testing	What problems can a too-long cable cause?	Attenuation, signal loss, data errors and slow speeds.	1	0/1	
17	2.3.5 Fault Detection	Name two common cable faults.	Any two: open, short, miscrossed wires, split pair, attenuation, crosstalk, wrong termination.	1	0/1	Named common cable faults
18	2.3.5 Fault Detection	What is a short circuit in a cable?	Two or more conductors touching or connected when they should not be.	1	1	
19	2.3.5 Fault Detection	What is an open circuit?	A break in a conductor, so no electrical path exists.	1	1	
20	2.3.5 Fault Detection	How does a TDR help locate a fault?	It sends a pulse down the cable and measures reflection time to calculate the distance to the fault.	1	1	Explained
21	2.3.6.1 Cable Tester	What is the function of a cable tester?	Checks continuity and wiring order of a cable.	1	1	
22	2.3.6.1 Cable Tester	What are the two parts of a basic cable tester?	The main unit and the remote unit.	1	1	
23	2.3.6.1 Cable Tester	How does a cable tester show results?	LED lights flashing in sequence, or a display showing pass/fail and fault type.	1	1	Explained
24	2.3.6.2 Multimeter	Name two measurements a multimeter can make.	Voltage, current, resistance, continuity (any two).	1	1	
25	2.3.6.2 Multimeter	How do you test continuity with a multimeter?	Set to continuity/resistance mode and touch the probes to both ends of the wire. A beep	1	1	

CamScanner

			or near-zero resistance means good continuity.			
26	2.3.6.3 Crimping Tool	What is a crimping tool used for?	Attaching connectors such as RJ-45 to the end of a cable.	1	01	Correct
27	2.3.6.3 Crimping Tool	Which connector is crimped onto a UTP Ethernet cable?	The RJ-45 connector.	1	01	given correctly
28	2.3.6.3 Crimping Tool	What should you do after crimping?	Test the cable with a cable tester to verify correct termination.	1	01	stated
29	2.3.6.4 Wire Stripper and Cutter	What is a wire stripper used for?	Removing the outer jacket or insulation without damaging the inner conductors.	1	00	
30	2.3.6.4 Wire Stripper and Cutter	What is the cutter used for?	Cutting cable to length and trimming wire ends evenly before termination.	1	01	
		Total		**30**	**/30**	22

Scoring guide: 27–30 Excellent | 22–26 Good | 15–21 Satisfactory | Below 15 Needs revision

6. Overall Assessment Summary

Assessment component	Maximum marks	Marks obtained
Practical observation checklist	50	49
Product checklist	5	4 1/2
Oral assessment checklist	30	22
Grand total	85	75 1/2

CamScanner

7. Competence Decision

Competence must be determined according to the institution's approved assessment criteria.

Assessment outcome	Tick
Competent-50 and above	✓
Not Yet Competent-49 and below	

Assessment Comments

Work on your oral Part!

Candidate Feedback

The oral part was a bit challenging.

CamScanner

Task 4

End A		End B
Pin -1	——→	-1 ✓
Pin -2	- - →	-2 ✓
Pin -3	- - →	-3 ✓
Pin -4	- - →	-4 ✓
Pin -5	- - →	-5 ✓
Pin -6	- - →	-6 ✓
Pin -7	- - →	-7 ✓
Pin -8	- - →	-8 ✓

08

The cable passed ✓

Straight through cable ✓

Task 6

Fault type : ~~Reversed pair~~ Split pair

Affected pins: Pin 1
Pin 2
Pin 3
Pin 6
Pin 8